

Programme	B. Sc. Mathematics Honours			
Course Code	MAT7CJ405			
Course Title	DISCRETE MATHEMATICS			
Type of Course	Major			
Semester	VII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	3	2	75
Pre-requisites	Basic Logical thinking and Set theory.			
Course Summary	The "Discrete Mathematics" course (MAT7CJ405) covers essential concepts in discrete structures and their applications. Students explore topics like graph theory, automorphisms, connectivity, and order relations through carefully structured modules. The course includes practical exercises and references to foundational works in the field, providing students with theoretical understanding and problem-solving skills necessary for further studies or real-world applications in mathematics and related areas.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe and explain fundamental concepts in graph theory, including subgraphs, vertex degrees, paths, connectedness, and operations on graphs.	U	C	Internal Exam/ Assignment/ Seminar/ Viva /Report/ End Sem Exam
CO2	Apply and analyse concepts related to automorphisms of graphs, vertex and edge cuts, and graph connectivity, utilizing definitions, theorems, and exercises.	An	P	Internal Exam/ Assignment/ Seminar/ Viva/ Report/ End Sem Exam
CO3	Evaluate and compare order relations in mathematical contexts and their implications for understanding and applying order theory.	E	C	Internal Exam/ Assignment/ Seminar/ Viva/ Report/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	1. A Textbook of Graph Theory. (2/e) Balakrishnan, R, & Ranganathan, K, Springer-Verlag, New York Inc., 2020 2. Foundations of Discrete Mathematics, K. D Joshi, New Age International (P) Limited, New Delhi, 1989. 3. An Introduction to Formal Languages and Automata (2/e), Peter Linz, Narosa Publishing House, New Delhi, 1997			
Module	Unit	Content	Hrs (75)	External Marks (70)
I	Fundamentals of Graph Theory		12	Min.15
	1	Section 1.0 Introduction (Text 1)		
	2	Section 1.1 Basic Concepts (Text 1)		
	3	Section 1.2 Sub Graphs (Text 1)		
	4	Section 1.3 Degrees of Vertices (Text 1)		
	5	Section 1.4 Paths and Connectedness (Text 1)		
II	Graph Operations and Connectivity		11	Min.15
	6	Section 1.5 Automorphisms of a simple graph (Definition 1.5.1 to Theorem 1.5.3) (Text 1)		
	7	Section 1.5 Automorphisms of a simple (Exercise 5.1 to Exercise 5.5) (Text 1)		
	8	Section 1.7 Operations on Graphs (Definition 1.7.1 to Example 1.7.10) (Text 1)		
	9	Section 1.7 Operations on Graphs (Exercise 7.3 to Exercise 7.6) (Text 1)		
	10	Section 3.1 Vertex Cuts and edge Cuts (Definition 3.1.1 to Theorem 3.1.10) (Text 1)		
	11	Section 3.1 Vertex Cuts and edge Cuts (Proposition 3.1.2 to Exercise 1.4) (Text 1)		
	12	Section 3.2 Connectivity and Edge - Connectivity (Definition 3.2.1 to Exercise 2.10) (Text 1)		
	13	Section 3.2 Connectivity and Edge - Connectivity (Theorem 3.2.10 to Theorem 3.2.11) (Text 1)		
III	Order Relations		11	Min.15
	14	Section 3 Order Relations (Sections 3, 3.1, 3.2 of Text 2)		
	15	Section 3 Order Relations (Sections 3.3, 3.4 of Text book 2)		
	16	Section 3 Order Relations (Sections 3.5, 3.6 of Text book 2)		
	17	Section 3 Order Relations (Sections 3.7 of Text book 2)		
	18	Section 3 Order Relations (Sections 3.8, 3.9, 3.10 of Text 2)		
IV	Finite Automata and Acceptors		11	Min.15
	20	Section 2.1 Deterministic Finite Acceptors (Text 3)		
	21	Section 2.2 Non-Deterministic Finite Acceptors (Text 3)		
	22	Section 2.3 Equivalence of Deterministic and Nondeterministic Finite Acceptors (Text 3)		

V	Practicum		30	
		Line Graphs and Directed Graphs		
		Eulerian Graphs and Hamiltonian Graphs		
		Planar and Non planar Graphs		
		Applications of Lattices in Switching Circuits		
		Applications of Automata in Theory of Computing		
References				
1. J. C. Abbot: Sets, lattices and Boolean Algebras; Allyn and Bacon, Boston; 1969. 2. J. A. Bondy, U.S.R. Murty: Graph Theory; Springer; 2000. 3. S. M. Cioaba and M.R. Murty: A First Course in Graph Theory and Combinatorics; Hindustan Book Agency; 2009 4. R. P. Grimaldi: Discrete and Combinatorial Mathematics- an applied introduction(5th edn.); Pearson; 2007. 5. J. L. Gross: Graph theory and its applications(2nd edn.); Chapman & Hall/CRC; 2005 6. Graph Theory and Decomposition, Jomon Kottarathil, Sudev Naduvath and Joseph Varghese Kureethara, CRC Press, London, New York, 2024.				

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	2	2	0	3	0	2	1	3	0	2
CO 2	1	3	2	1	3	0	3	2	3	0	3
CO 3	0	2	2	1	3	0	3	1	3	0	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/ Report	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓